Reproducible ML Portfolio | Actual Tesla T4 + CPU Results

Target: ML Engineering / AI Model Evaluation, Thomson Reuters. **Purpose:** measure accuracy, checkpoint size, and inference timing with a constrained compute budget. Do not mistake heterogeneous-device timing differences for quantization acceleration.

Experiment & provenance

Real scikit-learn/UCI handwritten digits dataset (1,797 examples); 1,347 train / 450 test; 64–128–64–10 MLP; three training seeds (11, 23, 37); 30 epochs/seed. 30 warmups + 120 timed single-item forward passes per variant/seed. 720 raw timing trials are included. Source: https://doi.org/10.24432/C50P49 (UCI CC BY 4.0).

Observed Colab execution (October 28, 2026)

Measurement	FP32 / Tesla T4 GPU	Dynamic INT8 / Colab CPU
Accuracy, mean ± SD	96.37% ± 0.68 pp	96.30% ± 0.79 pp
p50 latency, mean ± SD	0.242 ± 0.046 ms	0.262 ± 0.060 ms
p95 latency, mean ± SD	0.352 ± 0.085 ms	0.395 ± 0.061 ms
Serialized checkpoint	71,637 bytes	22,591 bytes

Hardware provenance: Tesla T4, CUDA available=True, PyTorch 2.10.0+cu128, Python 3.13.16, Colab host Intel Xeon @ 2.00GHz (2 logical CPUs). Hardware file: `results/hardware.json`. Uncertainty = between-seed sample SD, not a confidence interval. 68.5% checkpoint reduction; average accuracy difference about 0.07 pp.

What the results do — and do not — show

FP32 inference and training actually ran on the **GPU**. Dynamic INT8 inference ran on the **CPU** because this PyTorch dynamic quantization path is CPU-specific. The two latency values do **not** establish a GPU precision speedup/slowdown. Checkpoint bytes are not peak memory. Tiny single-input kernels, shared cloud scheduling, three seeds, and a digit-only dataset limit generalization.

Reproducibility & growth

Included: source code, pinned requirements, raw CSVs, hardware log, and a plot regenerated with `python src/experiment.py --mode plot`. Original independent CPU-only experiment archived in `results/cpu_baseline/` and never mixed with Colab results. Skills: profiling, uncertainty reporting, reproducibility, model documentation, hardware-aware interpretation.

Next experiment & business relevance

Repeat with two GPU-supported precision formats on the **same GPU**, plus larger models and more seeds. For Thomson Reuters, the transferable skill is evidence-based ML evaluation and auditable limitations, not digit accuracy as a proxy for legal AI quality.